

Question			Marking details	Marks available																																							
				AO1	AO2	AO3	Total	Maths	Prac																																		
9	(a)	(i)	First (pointing) finger points from {N → S / left to right / direction of the field} (1) Thumb points {in direction of motion [of AB] / up} (1) Second (middle) finger gives [induced] current which is <u>A to B</u> (1)	1 1	1		3																																				
		(ii)	<table border="1"><thead><tr><th rowspan="2">Change</th><th colspan="3">Effect on voltage</th><th colspan="3">Effect on time for 1 cycle</th></tr><tr><th>Decreases</th><th>Stays the same</th><th>Increases</th><th>Decreases</th><th>Stays the same</th><th>Increases</th></tr></thead><tbody><tr><td>Weaker magnets</td><td>✓</td><td></td><td></td><td></td><td>✓</td><td></td></tr><tr><td>Coil spins faster</td><td></td><td></td><td>✓</td><td>✓</td><td></td><td></td></tr><tr><td>More turns in the coil</td><td></td><td></td><td>✓</td><td></td><td>✓</td><td></td></tr></tbody></table> 1 mark for each correct row	Change	Effect on voltage			Effect on time for 1 cycle			Decreases	Stays the same	Increases	Decreases	Stays the same	Increases	Weaker magnets	✓				✓		Coil spins faster			✓	✓			More turns in the coil			✓		✓			3		3		
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	(b)	(i)	An <u>alternating</u> magnetic field (1) is linked to the secondary coil by the iron core (1) OWTTE	2			2																																				
		(ii)	Substitution: $\frac{25\,000}{400\,000} = \frac{600}{N_2}$ (1) Rearrange equation (1) So secondary turns = 16 × 600 = 9600 (1) Award 1 mark for answer of 16 or 0.0625	1	1 1		3	3																																			
			Question 9 total	5	6	0	11	3	0																																		